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DEPARTMENT OF DEFENSE HANDBOOK

APPLICATION AND SELECTION GUIDE/REFERENCE FOR CONTACTS AND FIBER OPTIC TERMINUS WHICH RELATE TO MIL-DTL-38999 ELECTRICAL CONNECTORS, AND WHICH ARE APPLICABLE TO MIL-STD-1560 INSERT ARRANGEMENT DIAGRAMS

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FOREWORD

Application and Selection Guide/Reference for Contacts and Fiber Optic Terminus which relate to MIL-DTL-38999 Electrical Connectors, and which are applicable to MIL-STD-1560 Insert Arrangement Diagrams.

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense

2. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply.

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1. SCOPE

1.1 Scope. This handbook provides history, background, supersession, guidance, and information on electrical contacts and fiber terminus allowed by MIL-DTL-38999 (Series I thru IV). In most cases, there are multiple electrical contacts (all conforming to AS39029) which may be used in a given cavity (identified by size – 23, 22D, 22M, 22, 20, 16, 12, 10, and 8). Connector configurations with contacts outside the range 23 – 8 and outside of MIL-DTL-38999 specifications are non-standard. MIL-STD-1560 defines the inserts and contact arrangements for all MIL-DTL-38999 connectors. Generally, MIL-DTL-38999 (vs. MIL-STD-1560) is used throughout this handbook. Contact information is or has been applicable to MIL-DTL-38999 and SAE-AS29600.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-32689	Connectors, Electrical, Circular, Miniature, Lightweight, High Density, Quick Disconnect, Tri-Start Threaded Coupling, Environment Resistant with Removable Crimp Contacts, General Specification for
MIL-PRF-29504/4	TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENTAL RESISTING, PIN TERMINUS, SIZE 16, REAR RELEASE, MIL-DTL-38999, SERIES III
MIL-PRF-29504/5	TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENTAL RESISTING, SOCKET TERMINUS, SIZE 16, REAR RELEASE, MIL-DTL-38999, SERIES III
MS27488	PLUG, END SEAL, ELECTRICAL CONNECTOR

(Copies these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein.

SAE INTERNATIONAL

SAE AS39029/27	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 12 (FOR MIL-DTL-38999 SERIES II CONNECTORS)
SAE AS39029/28	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, SHIELDED, SIZE 12 (FOR MIL-DTL-38999 SERIES I, II, III AND IV CONNECTORS)
SAE AS39029/56	CONTACTS, ELECTRICAL CONNECTOR, SOCKET CRIMP REMOVABLE (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/57	CONTACTS, ELECTRICAL CONNECTORS, SOCKET CRIMP REMOVABLE (FOR MIL-DTL-24308, MIL-DTL-38999 SERIES II, MIL-DTL-55302/68, /71, /72, /75 AND MIL-DTL-83733 CONNECTORS)

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SAE AS39029/58	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE (FOR MIL-DTL-38999 SERIES I, II, III, AND IV, AND MIL-DTL-55302/69 AND MIL-DTL-83733 CONNECTORS)
SAE AS39029/59	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 8 (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/60	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, SHIELDED, SIZE 8 (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/71	CONTACTS, ELECTRICAL CONNECTOR, PIN, REMOVABLE, SOLDERLESS WRAPPOST TERMINATION (FOR MIL-DTL-38999 SERIES I, II, III, AND IV, MIL-DTL-24308, MIL-DTL-83733, AND MIL-DTL-55302/69 CONNECTORS)
SAE AS39029/72	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, REMOVABLE, SOLDERLESS WRAPPOST TERMINATION (FOR MIL-DTL-38999 SERIES I, II, III, AND IV, MIL-DTL-24308, AND MIL-DTL-55302/68, /71, AND /75 CONNECTORS)
SAE AS39029/75	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 12 (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/76	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, SHIELDED, SIZE 16 (FOR MIL-DTL-38999 SERIES I, II, III, IV CONNECTORS AND 50 OHM CABLE)
SAE AS39029/77	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 16 (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/78	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 16 (FOR MIL-DTL-38999 SERIES II CONNECTORS)
SAE AS85049/80	CONNECTOR ACCESSORIES, ELECTRICAL, DUMMY CONTACT, SIZES 16, 12, AND 8, CATEGORY 7 (FOR MIL-DTL-38999 CONNECTORS)
SAE AS85049/81	CONNECTOR ACCESSORIES, ELECTRICAL, SEAL PLUG, SIZE 10, CATEGORY 7 (FOR MIL-DTL-38999 CONNECTORS)
SAE AS39029/87	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, THERMOCOUPLE (FOR MIL-DTL-38999 SERIES I, II, III, AND IV, MIL-DTL-24308, AND MIL-DTL-83733)
SAE AS39029/88	CONTACTS, ELECTRICAL CONNECTOR SOCKET, CRIMP REMOVABLE, THERMOCOUPLE (FOR MIL-DTL-38999 SERIES I, III, IV CONNECTORS)
SAE AS39029/89	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE (THERMOCOUPLE FOR MIL-DTL-38999 SERIES II, MIL-DTL-83733, AND MIL-DTL-24308 CONNECTORS)
SAE AS39029/90	CONTACT, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, PIN, SIZE 8
SAE AS39029/91	CONTACT, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, SOCKET, SHIELDED, SIZE 8
SAE AS39029/102	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, COAXIAL, SIZE 12 (FOR MIL-DTL-38999 SERIES I, II, III AND IV CONNECTORS)
SAE AS39029/103	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, COAXIAL, SIZE 12 (FOR MIL-DTL-38999 SERIES I, III AND IV CONNECTORS)

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SAE AS39029/106	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, (FOR MIL-DTL-38999 SERIES I, III, IV, MIL-DTL-55302 AND AS29600 SERIES A CONNECTORS)
SAE AS39029/107	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE (FOR MIL-DTL-38999 SERIES I, III, IV, MIL-DTL-55302, AND AS29600 SERIES A CONNECTORS)
SAE AS39029/113	CONTACT, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, PIN, SIZE 8 (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/114	CONTACT, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, SOCKET, SIZE 8, MIL-DTL-38999 SERIES I, III, AND IV
SAE AS39029/117	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, ENGINE APPLICATION, (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/118	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, ENGINE APPLICATION (FOR MIL-DTL-38999 SERIES I, III, AND IV CONNECTORS)
SAE AS39029/119	CONTACT, ELECTRICAL CONNECTOR, DIFFERENTIAL QUADRAX, PIN, SIZE 8, 100Ω, FOR MIL-DTL-38999 SERIES I, III, & IV CONNECTORS.
SAE AS39029/120	CONTACT, ELECTRICAL CONNECTOR, DIFFERENTIAL QUADRAX, SOCKET, SIZE 8, 100Ω, FOR MIL-DTL-38999 SERIES I, III, & IV CONNECTORS.
SAE AS39029/121	CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, EXPANDED PERFORMANCE, SIZES 10, 12, 16, 20, 22 AND 23 (FOR MIL-DTL-38999 SERIES III CONNECTORS)
SAE AS39029/122	CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, EXPANDED PERFORMANCE, SIZES 10, 12, 16, 20, 22 AND 23 (FOR MIL-DTL-38999 SERIES III CONNECTORS)

(Copies of these documents are available online at https://www.sae.org.)

3. DEFINITIONS

3.1 Definitions. The terms used in this standard are generally accepted by the electrical and electronics industries and commonly used in electrical connector engineering practice.

BIN	See “Basic Identification Number (BIN)”
Basic Identification Number (BIN)	A three number suffix that is assigned and indicated by a color code system (applied to the contact wire barrel for physical identification) to each individual contact in the AS39029 Specifications.
Co-ax contact	A coax contact is for a single wire with a shielded/ground braid. It does maintain impedance.
Contact	Individual electrical connection terminating a wire within a connector and interfacing with a corresponding contact in another connector when mated.
Contact size	Assigned numbers denoting the size of the contact mating end and the size of the contact wire barrel end.
Crimp contacts	A contact where a stripped wire is insert into the contact's barrel and the barrel is crimped (mechanically deformed) to mechanically and electrically join the wire and contact

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Dummy plug	Dummy plugs are used to fill unused insert/grommet cavities in a connector. There primary function is to seal the cavity. They also provide an expected force on the grommet. This force buttresses against adjacent contact. This aids in better sealing between the grommet and adjacent contacts. They are also referred to as sealing plugs.
Enhanced performance contacts	These are different 39029 contacts which were developed with enhanced performance above and beyond the standard 38999 contact.
Fiber contacts	Fiber optic terminus can be used with MIL-DTL-38999 connectors and will generally be specified as MIL-PRF-29504
Pin (contact)	A contact having a mating end that enters the socket contact.
Power & signal contact	See “Standard (power & signal) contact”
Quadrax	A single size 8 contact assembly consisting of 4 smaller contacts within. Used for high speed data transmission. 100 Ω impedance.
Shielded contact	A shielded contact is for a single wire with a shielded/ground braid. It does not maintain impedance.
Sealing plug	See “Dummy plug”
Standard (power & signal) contact	This type of contact is the traditional, historic contacts used in MIL-DTL-38999 connectors
Socket (contact)	A contact having a mating end that will accept entry of a pin contact.
Twinax	A twin-ax contact is for twisted pair (2 wires) with a shielded/ground braid. It does maintain impedance.
Thermocouple contacts	Thermocouple contacts are intended for use primarily for high temperature instrument probes and sensing devices; and providing connection between two in-line thermocouple wires.
Wrap post contact	Contact were the wire-contact electrical connection is made by helically wrapping the wire around a "post" feature on the rear of the contact. The wire is uninsualted (bare).

4. GENERAL INFORMATION

4.1 This section is not used.

5. DETAILED INFORMATION

5.1 Contacts and fiber optic terminus. A high level overview of different MIL-DTL-38999 contacts are provided in table I:

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TABLE I. High level overview of contacts for MIL-DTL-38999.

Type	SAE AS 32029 Slash Sheet	Pin or Socket	Contact sizes (mating)	Series				Details
				I	II	III	IV	
Standard (power & signal)	/58	Pin	10-22	■	■	■	■	Default supplied with 38999
	/56	Socket		■		■	■	
	/57				■			
Enhanced performance	/107	Pin	10-22	■		■	■	1500x mating cycles
	/106	Socket		■		■	■	
	/118	Pin	10-22	■		■	■	Hi-temp & vib. Requirements
	/117	Socket		■		■	■	
	/122	Pin	10-23			■		Vib. & shock requirements, expanded to size 23
	/121	Socket				■		
Dummy plugs	see detail	Both	0-23	■		■	■	specified by MS27488, no series II, redundant SAE options below
			8,12,16	■		■	■	specified by SAE AS85049/80, no series II
			10	■		■	■	specified by SAE AS85049/81, no series II
Shielded	/59	Pin	8	■		■	■	
	/60	Socket		■		■	■	
	/28	Pin	12	■	■	■	■	
	/75	Socket		■		■	■	
	/27				■			
	/76	Pin	16	■	■	■	■	
	/77	Socket		■		■	■	
	/78				■			
Co-ax	/102	Pin	12	■	■	■	■	no series II socket
	/103	Socket		■		■	■	
Twin-ax	/113	Pin	8	■		■	■	
	/114	Socket		■		■	■	
	/90	Pin		■		■	■	recommended for replacement purposes only, /113, /114 recommended for new design
	/91	Socket		■		■	■	
Quadrax	/119	Pin	8	■		■	■	Work in progress with SAE at time of this publication
	/120	Socket		■		■	■	

TABLE I. High level overview of contacts for MIL-DTL-38999. – continued

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Type	SAE AS 32029 Slash Sheet	Pin or Socket	Contact sizes (mating)	Series				Details
				I	II	III	IV	
Fiber	see detail	Pin	16					proposed only, specified by MIL-PRF-29504/4
		Socket						proposed only, specified by MIL-PRF-29504/5
Thermocouple	/87	Pin	16-22					
	/88	Socket						
	/89							
Wrap post	/71	Pin	22					helical wire wrap connection to contact
	/72	Socket						

Notes
1. Contacts are crimped unless otherwise noted
2. Some additional details are easily available in contact's document title in the Applicable Documents section of this handbook. For detail information see the individual specification.

5.1.1 Types. Further description and background is provided on the different types of contacts below. Unless otherwise stated, the types are copper connections (i.e. not fiber optic).

5.1.1.1 Standard (power & signal). This type of contact is the traditional, historic contacts used in MIL-DTL-38999 connectors. They are the original contacts for MIL-DTL-38999 Series I, II, III, and IV connectors and are the default contacts shipped with a MIL-DTL-39999 connector unless otherwise specified. They used the standard P and S designation in the part number. These contact specifications were converted from military specification MIL-C-39029/56, /57, and /58 respectively. They are the most commonly used contacts.

5.1.1.1.1 Short socket with Series II. Contacts specified in AS39029/57 are short socket contacts for MIL-DTL-38999 Series II connectors. Shorter contacts are needed for clearance. Since the standard pins (/58) is shorter than the standard socket (/56) and fit in series II, the pin contact for MIL-DTL-38999 series II connectors is the same as MIL-DTL-38999 Series I, III, & IV connectors.

5.1.1.2 Enhanced performance. These are different 39029 contacts which were developed with enhanced performance above and beyond the standard 38999 contact. Further information can be found in AS39029 base standard with Appendix A and the detail standards.

5.1.1.2.1 1500x mating cycles. These contacts were added to provide the 1500 mating and unmating cycle durability required for AS29600 and MIL-DTL-38999 composite connectors

5.1.1.2.2 Hi-temp & vibration. These contacts were added to provide increased durability and performance in high vibration and temperature environments (such as engine wiring). M39029/117 and /118 contacts have separate BIN codes for different Plating Options (Gold and Rhodium)

5.1.1.2.3 Vibration & shock. These contacts were added to align/harmonize the contact performance requirements with the applicable connector performance requirements and include additional verification testing for vibration and shock requirements, and to add size 23 contacts to support new insert arrangements (See MIL-STD-1560) for high density/small wire applications in MIL-DTL-38999 Series III. Size 23 contacts were new to MIL-DTL-38999 in the late 2010s.

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5.1.1.3 Dummy plugs. Dummy plugs are used to fill unused insert/grommet cavities in a connector. There primary function is to seal the cavity by keeping moisture from entering the connector rear sealing grommet. They also provide an expected force on the grommet. This force buttresses against adjacent contact. This aids in better sealing between the grommet and adjacent contacts. They are also referred to as sealing plugs. They are color coded for easy size identification. Sealing plugs are to be inserted by hand, large-end pointing toward the connector rear.

5.1.1.4 Shielded. A shielded contact is for a single wire with a shielded/ground braid. It does not maintain impedance. The construction of the conductor consists of a center conductor (signal) with an outer cylindrical sleeve (shield, ground). It differs in principal from a co-ax connector only in that does not control impedance. Concentricity in the design means the contact does not have to be install in an particular orientation in the insert/connector (which is an issue in some contacts like Quadrax).

5.1.1.4.1 /76, /77, /78 viability. BIN 424, 428, and 432 (RG316 cable) and 425, 429, and 433 (RG178 cable) are still viable at time of publication of this document. Other BIN codes not recommended for use. No known standard cable exist to replace the previously recommended commercial cables or the cables are no longer manufactured. Standard tools may not apply. Follow contact supplier's instructions.

5.1.1.5 Co-ax. A coax contact is for a single wire with a shielded/ground braid. It does maintain impedance. The construction of the conductor consists of a center conductor (signal) with an outer cylindrical sleeve (shield, ground). It differs in principal from a shielded connector only in that does control impedance. Concentricity in the design means the contact does not have to be install in an particular orientation in the insert/connector. Concentricity in the design means the contact does not have to be install in an particular orientation in the insert/connector (which is an issue in some contacts like Quadrax).

5.1.1.5.1 Unwired contacts. It is inappropriate to install unwired coax, twinax, quadrax contacts or fiber optic terminus in connectors due to the multi-piece design of the contacts/terminus. When cavities are not populated with contacts, use dummy plugs:

5.1.1.6 Twin-ax. A twin-ax contact is for twisted pair (2 wires) with a shielded/ground braid. It does maintain impedance. The construction of the conductor consists of a center conductor (signal #1), an intermediate cylindrical sleeve (signal #2), and an outer cylindrical sleeve (shield, ground). Concentricity in the design means the contact does not have to be install in an particular orientation in the insert/connector. Concentricity in the design means the contact does not have to be install in an particular orientation in the insert/connector (which is an issue in some contacts like Quadrax).

Contacts conforming to AS39029/90, /91, /113, and /114 are defined as concentric, and orientation latching into the connector cavity is not needed.

5.1.1.6.1 Unwired contacts. It is inappropriate to install unwired coax, twin-ax, quadrax contacts or fiber optic terminus in connectors due to the multi-piece design of the contacts/terminus. When cavities are not populated with contacts, use dummy plugs:

5.1.1.6.2 /90 and /91 consideration. SAE AS39029/90 and /91 contacts are recommended for replacement only. For new design /113 and /114 are recommended. The shift away from /90 and /91 is due to the mating geometry of the design. The center pin can be damaged by mismating and mishandling.

There are two design configurations specified and qualified currently for M39029/90-529 and M39029/91-530. The mating ends and essential external dimensions/features are the same for both contact designs. However, each design has unique strip lengths for the wire, and unique assembly tools. AS39029/90 and AS39029/91 provide figures and information for identification of the contact configuration, and the related

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assembly tools by part number.

5.1.1.7 Quadrax. These contacts are to be installed in the size 8 keyed contact cavity and consist of 4 mating pin/socket contacts in each Quadrax assembly. The contacts are crimped to high speed data cable (star quad ethernet). Nominal characteristic impedance = 100 Ω.

5.1.1.7.1 Unwired contacts. It is inappropriate to install unwired coax, twinax, quadrax contacts or fiber optic terminus in connectors due to the multi-piece design of the contacts/terminus. When cavities are not populated with contacts, use dummy plugs:

5.1.1.7.2 Installation. Quadrax contacts require alignment when installed in the connector insert. Connector insulators must have alignment keyway in the cavity to perform alignment during installation. For example standard MIL-DTL-38999 standard connectors do not have an alignment feature, therefore equivalent MIL-DTL-38999 non-QPL connectors must be used for these contacts, follow the supplier's instructions.

5.1.1.8 Fiber. Fiber optic terminus can be used with MIL-DTL-38999 connectors and will generally be specified as MIL-PRF-29504.

5.1.1.9 Thermocouple. Thermocouple contacts are intended for use primarily for high temperature instrument probes and sensing devices. They should be installed as defined by the instrument documentation. Thermocouple contacts consist of special materials (Alumel, Chromel, etc). They must be matched to the wire to which they were designed to be terminated. They are terminated with the same tooling as the corresponding copper contacts, however, the crimp force required to terminate these contacts/wires is higher than that for copper contacts and wire due to the hardness of the contact and wire materials. It may be advisable to use pneumatic crimp tools conforming to AS22520/28 or /29 for these crimp terminations.

5.1.1.10 Wrap post. Contact were the wire-contact electrical connection is made by helically wrapping the wire around a "post" feature on the rear of the contact. The wire is uninsualted (bare). An example is shown in figure 1.

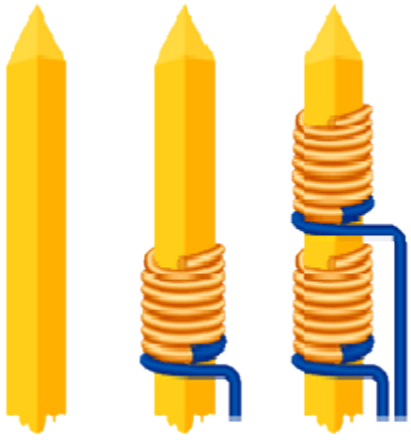


FIGURE 1. Depiction of wrap post with no, one, and two wire wraps.

5.1.2 Contact sizes. Contact size ranged from 10 to 22 historically. Size 23 contacts were new to MIL-DTL-38999 in the late 2010s.

5.1.2.1 Size 22. Size 22D is the preferred contact in this size category. Sizes 22 and 22M are for

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replacement only, and not recommended for new design

5.1.2.2 Size 23. Size 23 is not interchangeable with size 22, 22D or 22M, and should be used only where specified by MIL-DTL-38999 and MIL-STD-1560.

5.1.3 Contact part numbers and BIN codes: The current specification(s) for all electrical contacts relating to MIL-DTL-38999 connectors is the AS39029 base document, and the AS39029/XXX detail standards. Basic Identification Number (BIN) is a three number suffix that is assigned and indicated by a color code system (applied to the contact wire barrel for physical identification) to each individual contact in the AS39029 Specifications. For contact size 16 and larger a three digit BIN number representing the color bands may be stamped onto the contact. The stamped digits is a minimum of 0.060 inch high. Location of BIN code digits is optional. The BIN contact summary (Chart) can be found as APPENDIX A of the AS39029 base document.

5.2 Installation tools (39029): The termination and installation/removal tools are defined in a table in the SAE AS39029 detail specifications and listed in MIL-HDBK-1646.

5.2.1 Unwired contacts. Special unwired contact removal tools are required and specified for contacts which are installed without wires attached in MIL-DTL-38999 connectors. This is typical of most rear release connectors in aerospace wiring systems which have unwired contacts installed. Special tools which release and grip the contact wire barrel are required to compensate for the absence of wire.

5.2.2 Related resources.

Qualification information on the connectors can be found at::
(https://landandmaritimeapps.dla.mil/programs/milspec/default.aspx)
Qualification information on the contacts can be found at: (https://www.navair.navy.mil/qpl/)
Qualified Product Database (QPD) information can be found at: (https://qpldocs.dla.mil/)
Copies of MIL-DTL Standards are available at:
(https://landandmaritimeapps.dla.mil/programs/QmlQpl/)
Copies of AS39029 and related contact standards are available at www.sae.org.

6. NOTES

6.1 Intended use. The purpose of this document is to identify applications, background, and supersession of contacts for MIL-DTL-38999 connectors.

6.2 Subject term (key word) listing.

Contacts
Socket
Pin

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2022-115)

Review activities:
Navy – EC, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:45Z
Check the source to verify that this is the current version before use.